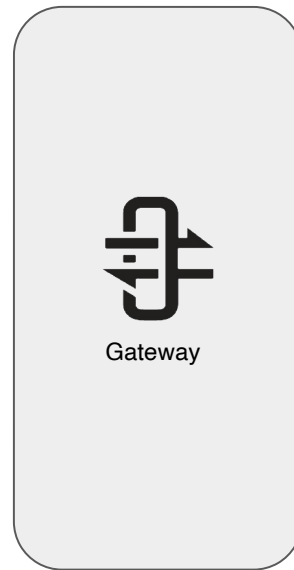


Tyk Architecture & Ecosystem

Understanding the main Tyk components, their functions and its dependencies

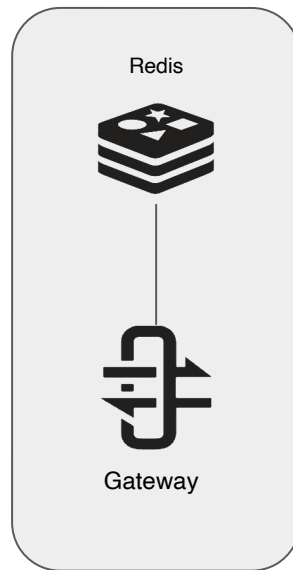
Tyk Gateway

- Real Time Traffic Enforcement
- Handles all inbound API traffic
- Authentication & Authorization
- Rate Limiting and Quotas
- Enforces Policies
- Observability



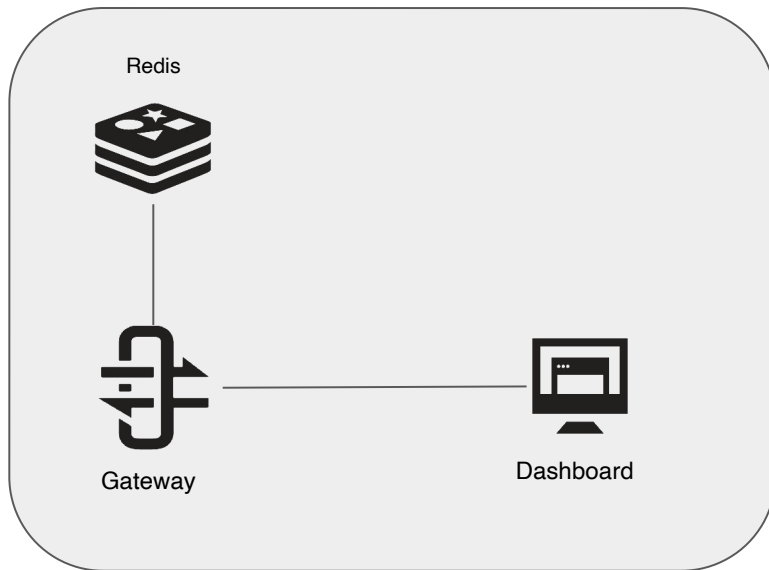
Tyk & Redis

- Fast in-memory performance
 - Enables real-time request validation
 - Minimise latency for high-throughput APIs
- Stores:
 - Rate limit counters
 - Session tokens
 - API key states
- Shared State across gateways
 - Synchronizes usage data across nodes
- Pub/Sub Broker
 - Cluster-wide notifications



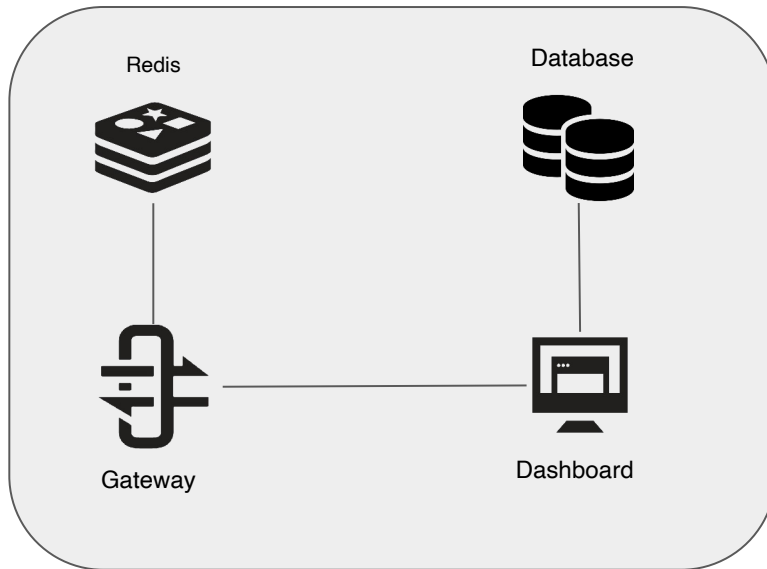
Tyk Dashboard

- Web-based Admin UI
 - Create, update, and organize your APIs visually
- Define access policies
 - Control who can access which APIs, how often, and with what credentials
- Manage users and teams
 - Set roles & permissions
- View real-time analytics
 - See traffic, errors, usage trends, and performance
- Push configuration to Gateway
 - Changes in the dashboard are synced automatically with Tyk Gateway
- Supports Single Sign On (SSO)



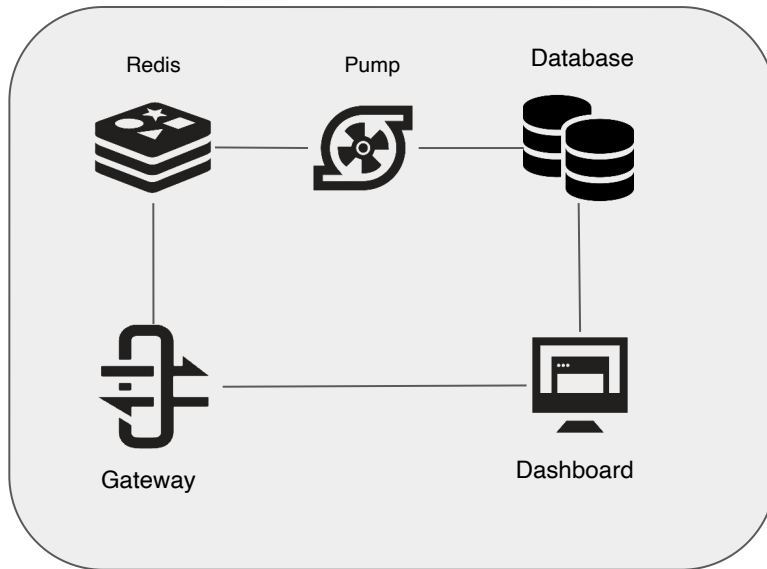
Database

- Dashboard data store
 - Configuration Storage
 - Analytics and Reporting
 - User and Access Management
 - Audit Logs



Tyk Pump

- Pumps analytics data to database
 - Data Aggregation
 - Data Transformation
 - Data Export
 - Flexibility



Tyk Pro

